

| Calculations                                                         |            |            |            |            |
|----------------------------------------------------------------------|------------|------------|------------|------------|
|                                                                      | Sheep      | Pig        | Cow        | Goat       |
| Meat Diet Portion                                                    | 3.35%      | 10.82%     | 10.30%     | 2.83%      |
| Meat MCal Target                                                     | 91,623.58  | 296,095.12 | 282,045.17 | 77,573.63  |
| Potential Dairy Value attached to each juvenile                      | 132.54     | 0.00       | 1015.20    | 126.90     |
| Value per juvenile                                                   | 48.00      | 160.00     | 640.00     | 48.00      |
| Adjusted Dairy Value attached to juvenile based on Dairy Utilization | 0.00       | 0.00       | 0.00       | 0.00       |
| Culled Mothers Value                                                 | 14.77      | 4.41       | 105.26     | 14.77      |
| Culled Adult Males                                                   | 0.74       | 0.22       | 5.26       | 0.74       |
| Final Value per juvenile                                             | 63.51      | 164.63     | 750.53     | 63.51      |
| Value MCals                                                          | 63.51      | 164.63     | 750.53     | 63.51      |
| Offspring required Year <sup>-1</sup>                                | 1,442.72   | 1,798.59   | 375.80     | 1,221.48   |
| MCals from meat                                                      | 91623.58   | 296095.12  | 282045.17  | 77573.63   |
| Max Utilized Mcals allowed from Dairy                                | 0.00       | 0.00       | 0.00       | 0.00       |
| MCals actually Utilized from Dairy                                   | 0.00       | 0.00       | 0.00       | 0.00       |
| Percent of diet from their meat                                      | 3.35%      | 10.82%     | 10.30%     | 2.83%      |
| Percent of Diet from their Dairy                                     | 0.00%      | 0.00%      | 0.00%      | 0.00%      |
| Kg Fleece                                                            | 10,902.60  | 0.00       | 0.00       | 0.00       |
| 1 Year+ (age) skins                                                  | 536.36     | 68.66      | 72.32      | 454.16     |
| Yearly Offspring skins                                               | 2,625.31   | 2,119.44   | 553.79     | 2,222.83   |
| Birthing Events per year                                             | 1.00       | 1.86       | 1.00       | 1.00       |
| Offspring Mother <sup>-1</sup> Year <sup>-1</sup>                    | 0.56       | 5.48       | 0.68       | 0.56       |
| Mothers Step 1                                                       | 2,576.28   | 328.51     | 552.64     | 2,181.22   |
| Males Step 1                                                         | 721.36     | 899.29     | 187.90     | 610.74     |
| Females Step 1                                                       | 721.36     | 899.29     | 187.90     | 610.74     |
| Replacement Mothers per year                                         | 515.56     | 66.00      | 69.27      | 436.54     |
| Breeding Males Required                                              | 135.25     | 17.25      | 29.01      | 114.51     |
| Additional Mothers                                                   | 2,111.78   | 58.60      | 261.76     | 1,788.12   |
| Males Step 2                                                         | 591.30     | 160.43     | 89.00      | 500.67     |
| Females 2                                                            | 591.30     | 160.43     | 89.00      | 500.67     |
| Total Mothers                                                        | 3,091.83   | 394.51     | 621.91     | 2,617.77   |
| Total Males                                                          | 1,312.66   | 1,059.72   | 276.90     | 1,111.41   |
| Total Females                                                        | 130.06     | 738.87     | 98.90      | 110.07     |
| Total Offspring                                                      | 2,625.31   | 2,119.44   | 553.79     | 2,222.83   |
| Total Animals                                                        | 4,669.81   | 2,210.35   | 1,026.72   | 3,953.76   |
| Units per Hectare                                                    | 1.00       |            |            |            |
| Total Adults per Hectare                                             | 1.41       | 0.22       | 0.35       | 1.41       |
| Total Offspring per Hectare                                          | 0.70       | 0.98       | 0.21       | 0.70       |
|                                                                      | 2.30%      |            | 6.47%      | 1.95%      |
| Total Adults                                                         | 5,339      | 470        | 913        | 4,520      |
|                                                                      |            |            |            |            |
|                                                                      |            |            |            |            |
| Total Offspring                                                      | 2,625      | 2,119      | 554        | 2,223      |
| Days for Maintenance                                                 | 5.00       | 220.00     | 0.00       | 5.00       |
| Dry Mothers                                                          | 466.52     | 0.00       | 68.12      | 394.94     |
| Other                                                                | 42,005.01  | 0.00       | 0.00       | 35,565.26  |
| MCals Dry Mothers                                                    | 544,897.40 | 0.00       | 144,208.01 | 461,286.81 |
| total Maintenance MCals                                              | 586,902.41 | 0.00       | 144,208.01 | 496,852.06 |

| MCals Required |             |             |             |             |
|----------------|-------------|-------------|-------------|-------------|
|                | Sheep       | Pig         | Cow         | Goat        |
| Gestating      | 1,563,374   | 967,632     | 927,046     | 1,323,694   |
| Lactating      | 2,982,618   | 343,986     | 996,823     | 2,525,355   |
| Maintenance    | 586,902     | 0           | 144,208     | 496,852     |
| Adult Males    | 157,977     | 0           | 61,422      | 133,753     |
| Offspring      | 720,648     | 2,130,042   | 2,026,874   | 610,166     |
| TOTAL          | 6,011,520.2 | 3,441,660.0 | 4,156,372.4 | 5,089,820.7 |

| Animal Diet Breakdown           |         |         |        |         |
|---------------------------------|---------|---------|--------|---------|
|                                 | Sheep   | Pig     | Cow    | Goat    |
| Crop                            | Barley  | Barley  | Barley | Barley  |
| Grams <sup>-1</sup> Straw/Chaff | 1000.0  | 0.0     | 4956.0 | 1000.0  |
| MCal kg <sup>-1</sup>           |         |         |        |         |
| Grams <sup>-1</sup> Grain       | 750.0   | 750.0   | 1500.0 | 750.0   |
| MCal kg <sup>-1</sup>           |         |         |        |         |
| Days                            | 0.0     | 0.0     | 0.0    | 0.0     |
| Number of Animals               | 2,625.3 | 2,119.4 | 553.8  | 2,222.8 |
| MCals from Straw/Chaff          | 0.0     | 0.0     | 0.0    | 0.0     |
| MCals from Grain                | 0.0     | 0.0     | 0.0    | 0.0     |
| MCals from other feed           | 0.0     | 0.0     | 0.0    | 0.0     |

| Diet Calculations                           |               |         |
|---------------------------------------------|---------------|---------|
| Plants Mcal                                 | 1,916,250.000 |         |
| Meat Mcal                                   | 821,250       |         |
| Dairy                                       | 0             |         |
|                                             | Perfect       |         |
| Plants %                                    | 70.00%        |         |
| Meat %                                      | 30.00%        |         |
| Dairy %                                     | 0.00%         |         |
| check                                       | Perfect       |         |
| Percent                                     |               |         |
| Domestic Meat                               | 27.30%        |         |
| Dairy                                       | 0.00%         |         |
| Wild Meat                                   | 2.70%         |         |
| Crops                                       | 63.00%        |         |
| Wild Plants                                 | 7.00%         |         |
| check                                       | Perfect       |         |
|                                             |               |         |
| new dairy                                   | 0.00          |         |
| potential dairy < limit                     | TRUE          |         |
| new limit                                   | 70.0%         |         |
| the limit greater than that left to satisfy | TRUE          |         |
| limit                                       | 1             |         |
| limit dairy checked                         | FALSE         |         |
| Mcalories                                   |               | Surplus |
| Domestic Meat                               | 747,337.500   |         |
| Dairy                                       | 0.000         |         |
| Wild Meat                                   | 73912.500     |         |
| Crops                                       | 1,724,625.000 |         |
| Wild Plants                                 | 191,625.000   |         |
| check                                       | Perfect       |         |
|                                             |               |         |
| SURPLUS DAIRY                               | 0.000         |         |
| This could account for additional % of diet | 0.00%         |         |

Herd Modelling

|                                     | Sheep   | Pig     | Cow     | Goat    |  |  |  |  |
|-------------------------------------|---------|---------|---------|---------|--|--|--|--|
| Mothers Req'd                       | 3092    | 395     | 622     | 2618    |  |  |  |  |
| Males for Slaughter                 | 1313    | 1060    | 277     | 1111    |  |  |  |  |
| Females for Slaughter               | 130     | 739     | 99      | 110     |  |  |  |  |
| Offspring available for Consumption | 2625    | 2119    | 554     | 2223    |  |  |  |  |
| Gestation Req'd                     | 859138  | 821146  | 629083  | 727394  |  |  |  |  |
| Lactation Req'd                     | 1639070 | 291911  | 676434  | 1387728 |  |  |  |  |
| Maintenance Req'd                   | 1949254 | 0       | 185360  | 1650401 |  |  |  |  |
| Adult Males                         | 157977  | 0       | 61422   | 133753  |  |  |  |  |
| Offspring Required                  | 862415  | 2320792 | 2425603 | 730199  |  |  |  |  |
| Energy Required                     | 5467854 | 3433848 | 3977902 | 4629474 |  |  |  |  |
| Ewe Fleece Kg Yr-1                  | 6957    |         |         |         |  |  |  |  |
| Ram Fleece Kg year-1                | 271     |         |         |         |  |  |  |  |
| Juvenile Fleece Kg Yr-1             | 3675    |         |         |         |  |  |  |  |
| Kg Fleece Year-1                    | 10903   | 0       | 0       | 0       |  |  |  |  |